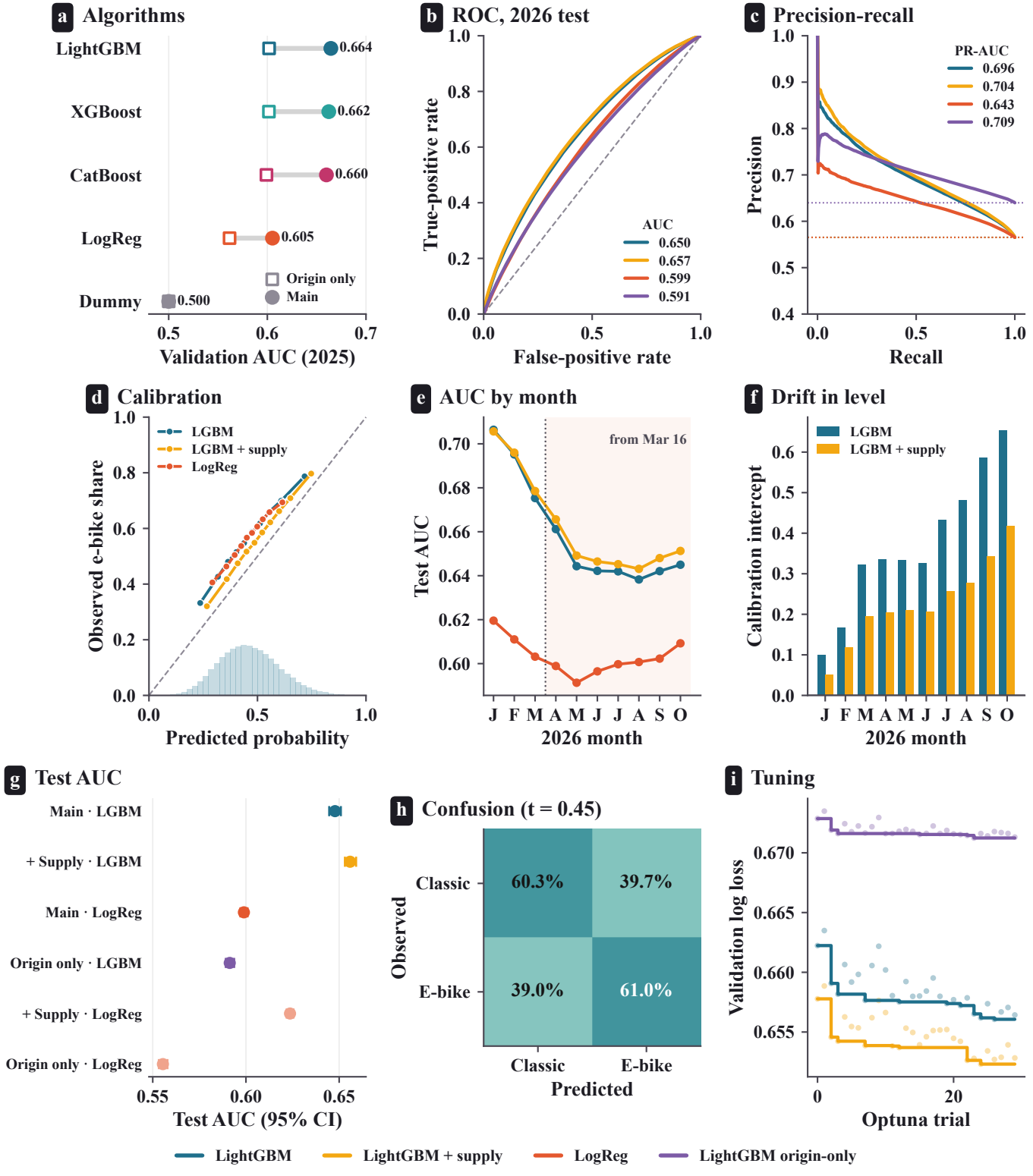
a Algorithms
LightGBM
XGBoost
CatBoost
LogReg
Dummy
0.664
0.662
0.660
0.605
0.500
Origin only
Main
0.5
0.6
0.7
Validation AUC (2025)
b ROC, 2026 test
True-positive rate
False-positive rate
AUC
0.650
0.657
0.599
0.591
c Precision-recall
Precision
Recall
PR-AUC
0.696
0.704
0.643
0.709
d Calibration
Observed e-bike share
Predicted probability
LGBM
LGBM + supply
LogReg
e AUC by month
Test AUC
from Mar 16
J F M A M J J A S O
2026 month
f Drift in level
Calibration intercept
LGBM
LGBM + supply
J F M A M J J A S O
2026 month
g Test AUC
Main · LGBM
+ Supply · LGBM
Main · LogReg
Origin only · LGBM
+ Supply · LogReg
Origin only · LogReg
0.55
0.60
0.65
Test AUC (95% CI)
h Confusion (t = 0.45)
Observed
Classic
E-bike
60.3%
39.7%
39.0%
61.0%
Classic
E-bike
Predicted
i Tuning
Validation log loss
Optuna trial
LightGBM
LightGBM + supply
LogReg
LightGBM origin-only